

FARM MANAGEMENT SYSTEM – PROJECT PLANNING DOCUMENT

1. Project Title

Farm Verse Precision Agriculture Management System

2. Problem Statement

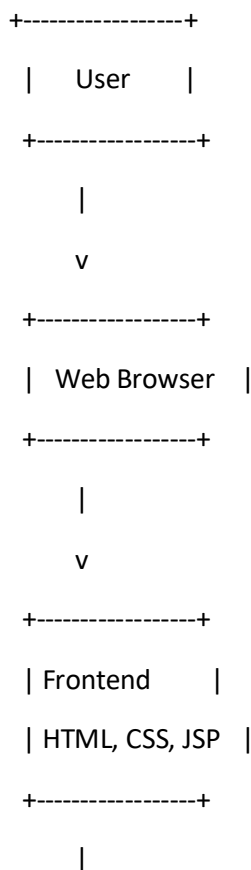
Farmers often maintain records of crops, livestock, expenses, and sales manually, which can lead to data loss, errors, and inefficient management. Traditional methods make it difficult to monitor farm activities, analyze, productivity, and make informed decisions. There is a need for a digital system that simplifies farm management and improves efficiency.

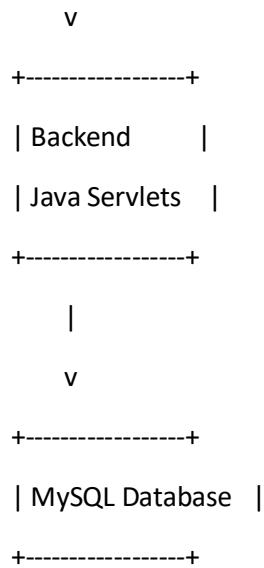
3. Proposed Solution

The Farm Management System is a web-based application designed to help farmers manage their daily farming activities efficiently. The system provides modules for user authentication, crop management, livestock management, expense tracking, sales management, and report generation. By digitizing farm records, the application reduces paperwork, improves data accuracy, and supports better decision-making.

4. Platform Architecture

The application follows a three-tier architecture:





Web Server:

Apache Tomcat

Version Control:

Git & GitHub

5. Technology Stack Component

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	Java Servlets
Database	MySQL
IDE	Eclipse IDE
Web Server	Apache Tomcat
Version Control	Git & GitHub

6. Functional Modules

- 1) User Registration and Login
- 2) Dashboard
- 3) Crop Management
- 4) Livestock Management
- 5) Expense Management
- 6) Sales Management
- 7) Report Generation
- 8) Logout

7. Project Objectives

- 1) Develop a user-friendly web application for farm management.
- 2) Maintain digital records of crops, livestock, expenses, and sales.
- 3) Improve data accuracy and reduce manual paperwork.
- 4) Generate reports to help users monitor farm activities.
- 5) Provide secure user authentication.

8. Expected Outcome

The Farm Management System will provide a centralized platform for managing farm operations digitally. Users will be able to maintain accurate records, track expenses and sales, monitor crop and livestock details, and generate reports. The system will improve efficiency, reduce manual effort, and support better farm management.

9. Future Enhancements

- 1) Weather forecasting integration
- 2) Mobile application support
- 3) AI-based crop recommendations
- 4) Online marketplace for farm products
- 5) SMS and Email notifications
- 6) Multi-user access with role-based permissions

10. Conclusion

The Farm Management System aims to modernize agricultural record management by providing a secure, efficient, and user-friendly web application. It will simplify farm operations, improve productivity, and enable better decision-making through digital record management.